DO NOW — RECALL

LMC instructions and data

Name: _____ Class: _____ Date: _____

1. What does direct addressing [30] retrieve?

2. What is the accumulator?

3. What is a mnemonic?

4. Search for 19 in sorted [1,4,7,10,13,16,19,22,25]. Show midpoint indexes checked.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The data at memory address 30.

2. Model answer

A CPU register holding an arithmetic/intermediate value.

3. Model answer

A readable abbreviation of an operation code.

4. Model answer

low0 high8 mid4→13, low5 high8 mid6→19; found index 6.

Vocabulary — know these terms

LMC — Little Man Computer teaching instruction set.

ACC — Accumulator holding current working value.

mailbox — Numbered LMC memory location.

STA — Store accumulator to memory.

LDA — Load accumulator from memory.

DAT — Declare/initialise a data location.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

LDA A restores the first input 9 and SUB B produces 5; OUT prints 5.

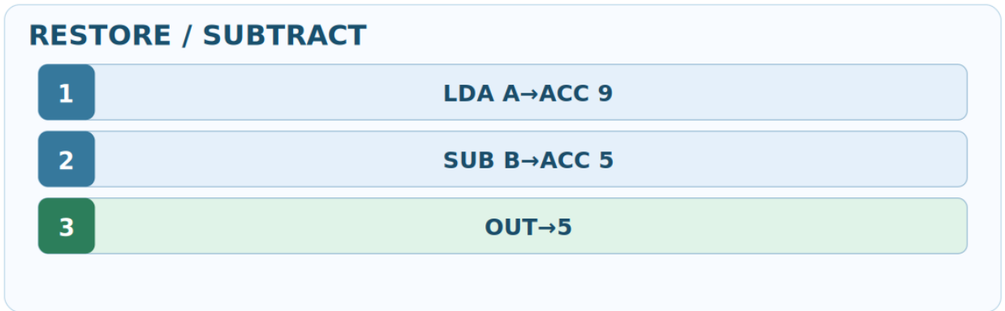
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Write and trace LMC code to calculate first input 9 minus second input 4.

Store input 9 as A and input 4 as B, since the accumulator is overwritten by INP.



RESTORE / SUBTRACT: LDA A→ACC 9; SUB B→ACC 5; OUT→5

NEW LEARNING

Read, annotate and explain

Know LMC's parts

LMC uses numbered mailboxes to hold instructions and data, and an accumulator for working values.

Trace instructions in order

First INP enters 4 into ACC, then STA FIRST copies 4 to the data mailbox before another input overwrites ACC.

Produce the result

Second INP makes ACC=7; ADD FIRST adds stored 4 to make 11, and OUT prints 11. HLT ends the program.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

LMC TRACE · INPUT 4 AND 7

INSTRUCTION	ACC / OUTPUT	FIRST MAILBOX
INP	4	0
STA FIRST	4	4
INP	7	4
ADD FIRST	11	4
OUT	prints 11	4

Store the first input before INP overwrites ACC with the second.

LMC trace: INP accumulator 4, STA FIRST stores 4, INP accumulator 7, ADD FIRST accumulator 11, OUT prints 11

New Learning

SECOND INPUT

1. INP: ACC=7
2. ADD FIRST: ACC=11

SECOND INPUT: INP: ACC=7; ADD FIRST: ACC=11

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In LMC, STA X replaces ACC with the stored value of X.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

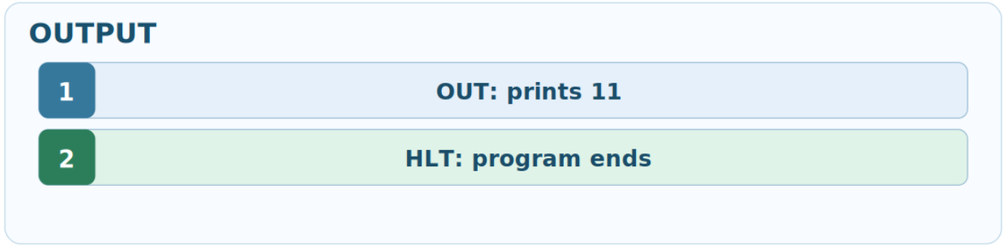
Question 3

Write an OCR LMC program that takes two inputs from 0 to 999, with the first greater than or equal to the second. Subtract the second from the first and output the non-negative result. Trace inputs 9 and 2.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



OUTPUT: OUT: prints 11; HLT: program ends

I DO — WORKED EXAMPLE

Follow the worked method

Write and trace an LMC program adding inputs 3 and 5.

Predict

Store the first input 3 into NUM so the second input will not overwrite it.

Explain

Read the second input 5, add NUM to get 8 and output the result.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain what INP, OUT, LDA and STA do in LMC.

Question 2

Trace INP; STA X; INP; ADD X; OUT; HLT; X DAT 0 using inputs 6 and 8. State each change and output.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict Store input 9 as A and input 4 as B, since the accumulator is overwritten by INP.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

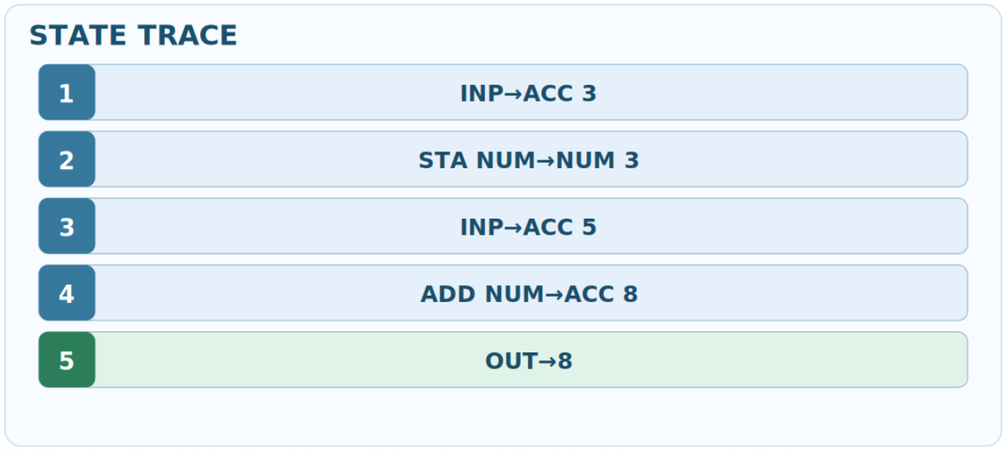
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Write and trace an LMC program adding inputs 3 and 5.

Store the first input 3 into NUM so the second input will not overwrite it.



STATE TRACE: INP→ACC 3; STA NUM→NUM 3; INP→ACC 5; ADD NUM→ACC 8; OUT→8

WE DO — GUIDED PRACTICE

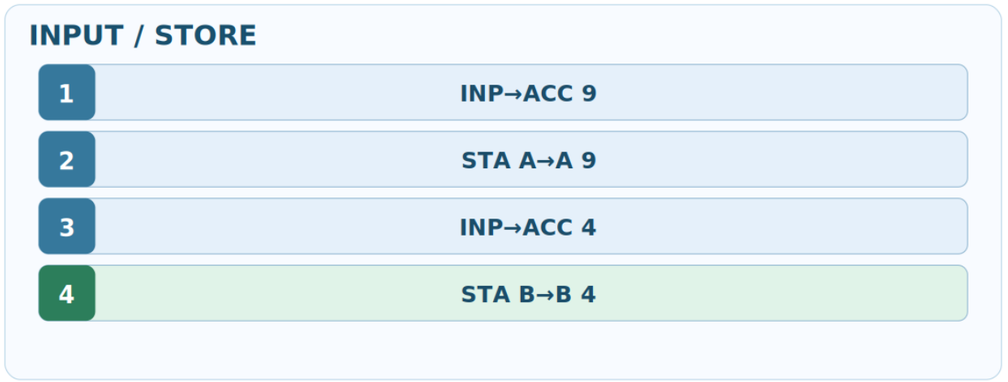
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Write and trace LMC code to calculate first input 9 minus second input 4.

Store input 9 as A and input 4 as B, since the accumulator is overwritten by INP.



INPUT / STORE: INP→ACC 9; STA A→A 9; INP→ACC 4; STA B→B 4